

REVISED TOPIC PROPOSAL

(Revision of Proposed Project No. 2)

Proposed Project No.	2 (Revised)
Proposed Title	BARANGAYAN: A Digital Barangay Management System Implementing DRRM, Medical Assistance, and Incident Reporting
Research Area and SDG	Research Area: • Mobile Computing, through cross-platform application development using React Native and Expo. • Electronic Governance, involving the digitization of local government services through a configurable system usable by multiple barangays. • Geographic Information Systems, applied through web-based mapping using Leaflet.js and OpenStreetMap data for incident and evacuation-site visualization. • Real-Time Data Management, implemented through Supabase's real-time database, authentication, and server-side functions. • Digital Financial Transactions, through the integration of PayMongo and QR Ph for online payment processing. Alignment with the Sustainable Development Goals (SDGs): • SDG 11 (Sustainable Cities and Communities): Target 11.3, which promotes inclusive and participatory planning, and Target 11.5, which aims to reduce the negative effects of disasters through timely information dissemination. • SDG 16 (Peace, Justice, and Strong Institutions): Target 16.6, which calls for the development of effective, accountable, and transparent institutions, supported in this project through measurable service tracking.
Problem Statement	In a considerable number of the Philippines' more than 42,000 barangays, essential government services—such as the issuance of barangay clearances, certificates of indigency, and residency certificates—continue to rely on manual, paper-based processes and in-person queuing. As a result, residents often experience long waiting periods without any indication of how long their request will take or whether it has reached the appropriate office for processing. In addition, reports of local incidents and hazards, such as flooding, road damage, and illegal dumping, are typically recorded on paper or shared informally through messaging applications. This makes it difficult for barangay officials to identify which areas require attention or to coordinate an appropriate response during emergencies such as typhoons or flooding. Most existing barangay digitization efforts are also designed for the specific needs of a single barangay, including its own list of documents, fees, and procedures. This limits their usefulness elsewhere, as adapting them to other barangays typically requires substantial redevelopment. These conditions demonstrate the need for a single, adaptable mobile platform that any barangay may use to digitize service requests, monitor processing times, visualize incident reports on a map, and share emergency information, without the need for separate development for each barangay.

Project Description	BARANGAYAN is a real-time electronic governance and community safety platform. It is developed using React Native and Expo, which allow the mobile application to run on both Android and iOS devices from a two separate codebase. The system uses Supabase as its backend, which provides the database, user authentication, real-time data updates, and server-side functions. A website version of the platform is also developed, and it connects to this same backend. Unlike existing barangay applications, which are typically built to serve only one barangay, BARANGAYAN is designed to be adaptable. Through an administrative setup feature, any barangay may define its own list of documents, fees, processing time targets, and incident categories, without requiring any changes to the program itself. This allows the same platform to be used by different barangays with only minor setup. BARANGAYAN is developed as a single system that can be accessed through two platforms: a mobile application and a website. The mobile application supports a resident account while the website supports a resident and an administrator account. This means that a resident may request documents, report incidents, or check the status of a request using either the mobile application or the website, whichever is more convenient. A barangay official may process requests, view the incident map, or post announcements using only the website platform. Since both platforms are connected to the same Supabase backend, any action performed on one platform is reflected on the other platform immediately. For example, if a resident submits a request through the website, an administrator will see this request right away when using the website. This ensures that residents and administrators always see the same, updated information. Payments for document and service fees are processed through PayMongo using QR Ph. This allows residents to pay using any participating bank or e-wallet application by scanning a single, standardized QR code.
Partner Organization / Target Users	Partner Organization (Pilot Site): - Barangay Ampid I, San Mateo, Rizal, which will serve as the initial partner for requirements validation and user testing. The administrative configuration feature of the system is designed so that other barangays may also adopt the platform independently, without requiring the specific setup used by the partner barangay. Target Users: Barangay Residents: individuals who request documents and services, report local incidents with supporting photographs, monitor the progress of their requests, receive community and emergency announcements, and check in during evacuation procedures. Barangay Officials and Administrators: personnel who configure the barangay's list of documents and processing time targets, manage resident records, process requests, update the status of incident reports, review geographic data on reported incidents, and issue emergency announcements.

Features (Consolidated Modules)

The features originally proposed have been reorganized into eight modules. No feature has been removed; related or overlapping items have been combined into shared components so that the system remains achievable within a six-month development period.

1. Digital Service and Document Request Module

- **Configurable document and service catalog:** allows each barangay's administrator to define its own document types, requirements, fees, and pickup schedules in a single interface. This replaces the separate, static "document admission guide" with detailed pages that update automatically when the administrator makes changes.
- **Shared appointment scheduling feature:** is used both for document pickup and for barangay health center appointments, such as check-ups, prenatal care, and vaccinations, with daily slot limits. Using one shared feature for both purposes avoids the need to build a separate scheduling system.
- **Resident profile management:** allows residents to maintain a secure profile containing household information and a valid identification document, which administrators may review and verify electronically.

2. Automated Service Tracking Module

- **A single, shared tracking feature:** is used to monitor the progress of documents, incident reports, and appointments, following a common sequence: submitted, in progress, and completed or resolved. Each type of request may have its own target processing time, and the system automatically flags requests that are near or beyond this target.
- **Real-time updates:** are provided through Supabase's real-time database, allowing residents to see changes in status immediately, without needing to refresh the application.

3. Community Incident Reporting and Mapping Module

- **Incident reporting by residents:** includes the ability to submit photographic evidence and select a category for the incident, such as a damaged light post, uneven road, blocked drainage, or illegal dumping, among other categories that may be added by the administrator.
- **An interactive map for administrators:** is built using Leaflet.js together with OpenStreetMap data, both of which are free to use and do not require a paid subscription. Reported incidents are grouped on the map according to category and resolution status.
- **Location verification during registration:** uses the applicant's device location as a preliminary check rather than an automatic requirement. Accounts registered from outside the barangay's boundaries are flagged for manual review by an administrator, in recognition of the limited accuracy of consumer-grade location data.

4. Emergency and DRRM Information Module

- **A categorized announcement feature:** delivers both routine community updates and emergency alerts through push notifications, with a text message option available for residents without access to mobile data.
- **A disaster preparedness guide and directory of emergency contacts:** which include first-aid instructions, evacuation checklists, and hotline numbers, are stored within the application so that they remain accessible even without an internet connection.
- **A directory of evacuation centers shown on the map, with in-app nearest-route display (Haversine Formula + OSRM Contraction Hierarchies / Dijkstra's Algorithm):** displays each center's location and its current occupancy status, as updated by an administrator. The resident's current device location is also shown

on the same map. Upon selecting a center, or automatically for the nearest center, the application calculates and displays the actual road route from the resident's current location to the evacuation center directly on the map, rather than requiring the resident to leave the application. A shortlist of alternative nearby centers and their routes is also available for the resident to compare.

- **QR code check-in linked to family status:** allows residents to scan a code upon arriving at an evacuation center, after which their status becomes visible to other family members registered under the same household account. This combines the previously separate check-in and family reunification features into one.

5. Digital Payment Module

- **Payment through PayMongo using QR Ph:** allows residents to pay document and service fees using a single, standardized QR code that is accepted by participating banks and e-wallet applications. The status of the corresponding request is updated automatically once payment is confirmed.

6. Administrative Analytics Dashboard

- Provides administrators with a real-time summary of total residents, requests grouped by type and status, the rate of compliance with processing time targets, and a summary of incident categories, in support of informed decision-making.

7. Account, Security, and Privacy Compliance

- Implements role-based access control and uses Supabase's server-side functions to manage administrative accounts securely, keeping sensitive account operations separate from the mobile application itself.
- Provides password recovery through a one-time password sent to the resident's registered contact information.
- Complies with the Data Privacy Act of 2012 (Republic Act No. 10173) by allowing residents to request a copy of their personal data or to have their account permanently deleted.

8. Help and Support Module

- Includes an expandable list of frequently asked questions, along with contact information such as a hotline number, email address, office hours, and physical address, to assist residents who require further help.

9. Trash and Illegal Dumping Mapping Module

- **Report grouping by area:** Trash-related and illegal-dumping reports, already collected through the Community Incident Reporting and Mapping Module, are grouped by location into small zones within the barangay, using the same incident data and categories already defined in that module.
- **Priority scoring (Time-Decay Weighted Frequency Scoring):** Each area receives a score based on two factors: how many reports it has received, and how recent those reports are, with recent reports weighted more heavily. This uses the same rolling, time-based scoring method already applied in the Automated Service Tracking Module, run through a scheduled Postgres function (pg_cron) on Supabase.
- **High-priority flagging:** Areas with the highest scores are marked as high priority on the administrator's incident map, allowing officials to identify areas with frequent, recent trash reports without reviewing each report individually.
- **Collection scheduling support:** Once an area is flagged, administrators may schedule trash collection ahead of the regular schedule, or dispatch collection immediately if the score indicates urgency. The system supports this decision; the final choice remains with the barangay official.

10. Medical and Vaccination Drive Module

Provides a better and efficient queueing that can automate how to allocate the vaccine and medicine and how to have a seamless vaccination process.

Resident side

- **Browse active drives** - medicine/vaccine type, date, venue, eligibility criteria, remaining stock (live-updating via Supabase Realtime)
- **Register** - submits basic eligibility info (age, PWD status, comorbidities, prior dose history for multi-dose vaccines); system auto-computes a priority score
- **Priority queue position** - resident sees their rank, not just "registered"
- **Status tracking** - registered → confirmed → dispensed, reusing your existing tracking states

Admin side

- **Create a drive** - medicine/vaccine name, total stock, date, venue, priority weighting rules
- **Priority queue dashboard** - residents auto-sorted by computed score; admin can see who's allocated vs. waitlisted
- **Live inventory** - remaining stock decrements in real time as QR codes are scanned/dispensed
- **Waitlist and no-show handling** - if a top-priority registrant no-shows by a cutoff time, their slot releases to the next-highest-priority waitlisted resident

RRLS LINK PER FEATURES:

1. Digital Service and Document Request Module

https://paraverse.feutech.edu.ph/repository/article/10.1007/978-981-19-7663-6_13

2. Automated Service Tracking Module

<https://www.scribd.com/document/698157062/EasyDocs-Barangay-Document-Request-System-FINALPaper>

3. Community Incident Reporting and Mapping Module

<https://dl.acm.org/doi/abs/10.1145/3537693.3537746>

4. Emergency and DRRM Information Module

<https://iopscience.iop.org/article/10.1088/1755-1315/97/1/012004>

<https://ieeexplore.ieee.org/document/10630716/>

5. Digital Payment Module

<https://www.bsp.gov.ph/Regulations/Issuances/2023/M-2023-005.pdf/>

<https://www.gmanetwork.com/news/money/economy/863474/bsp-mandates-full-adoption-of-philippine-qr-code-standard-by-july-1/story/>

6. Administrative Analytics Dashboard

<https://www.warse.org/IJATCSE/static/pdf/file/ijatcse071042021.pdf>

7. Account, Security, and Privacy Compliance

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4582212

8. Help and Support Module

<https://ieeexplore.ieee.org/document/11091916>

9. Trash and Illegal Dumping Mapping Module

https://doi.org/10.5194/isprs-archives-XLVIII-5-W4-2025-47-2026_2026_/

<https://isprs-archives.copernicus.org/articles/XLVIII-5-W4-2025/47/2026/isprs-archives-XLVIII-5-W4-2025-47-2026.pdf>

<https://ijsra.net/content/waste-disposal-knowledge-attitude-and-practices-barangay-poblacion-compostela-cebu>

10. Medical and Vaccination Drive Module

[Frontiers | A decision support system for the optimal allocation and distribution of COVID-19 vaccines using analytic hierarchy process \(AHP\) and integer programming \(IP\) model](#)

- The use of weighted prioritization scoring under supply constraints is supported by Culaste et al. (2023), who developed an AHP-derived weighted scoring system combined with integer programming to allocate COVID-19 vaccines across Philippine communities under limited supply and station capacity, deferring lower-priority allocations when capacity is exhausted. Unlike that community-level model, BARANGAYAN applies weighted scoring at the individual resident level, using a priority queue data structure for allocation rather than a linear programming solve a lighter-weight approach appropriate for a single-barangay, real-time mobile context rather than a multi-site, multi-day municipal rollout."
 - **Weighted eligibility scoring** → their AHP-derived weights w_{ij}
 - **Limited, exhaustible supply forcing prioritization** → their vaccine/capacity constraints
 - **Lower-priority parties deferred when supply runs out** → their Day 1 → Day 2 carryover

Novelty of the Project

Using the five core elements of novelty as a framework, and comparing BARANGAYAN against the four related studies provided, the analysis below identifies where BARANGAYAN's novelty is strong, where it overlaps with existing work, and where the project's contribution should be framed carefully.

1. Feature Integration (Strongest Novelty Claim)

Definition applied: Combining two or more separate tools or systems into one connected platform.

The four related studies each address only one function of barangay or local government operations:

- **REPORTIT:** focuses only on crime incident reporting and mapping for the police, developed for a single city (Calamba).
- **LBigayAksyon:** focuses only on emergency and accident response coordination between residents and rescue personnel, developed for a single municipality (Los Baños).
- **The Paracale study:** evaluates a general e-governance platform for one LGU, focusing on system quality (functionality, usability, security) rather than combining multiple service types.
- **Webyu (Digital Barangay):** focuses only on records management and administrative workflow automation for one barangay (Yuson), through a web-based system without a mobile component.

Each of these systems was built to solve one problem for one local government unit. “BARANGAYAN's clearest novelty is that it combines four functions that the literature treats as separate systems” — document and service request processing, incident and hazard reporting with mapping, emergency and evacuation coordination, and digital payment — into a single platform used by both residents and officials. This directly addresses a gap the literature demonstrates: local governments currently need several disconnected systems to cover what BARANGAYAN provides in one.

2. New Application

Definition applied: Applying an existing technology or method to a setting where it has not been applied before.

- None of the four studies apply a “configurable, reusable system design” to the barangay setting. REPORTIT, LBigayAksyon, and Webyu were each built for one specific local government unit, and would require redevelopment to be used elsewhere. The Paracale study evaluates an existing platform rather than proposing a model that other LGUs could adopt directly.
- BARANGAYAN applies a “configuration-based design approach” where an administrator defines document types, fees, and processing time targets without requiring new code to the barangay governance setting. This design pattern is common in other software industries (for example, business management platforms used by multiple companies), but its application to Philippine barangay governance, where the reviewed literature shows only single-barangay systems, represents a new application of this approach in this specific setting.

3. Improved Efficiency

Definition applied: Making an existing process faster, more accurate, or easier to monitor.

- Webyu reports that it "reduced processing times" through workflow automation, and the Paracale study notes that response time is an area identified for improvement in existing e-governance systems. However, neither study describes an “ automated compliance-tracking mechanism” that flags a request as delayed without manual checking.
- BARANGAYAN's Automated Service Tracking Module improves on this by automatically comparing the time elapsed against a defined target and flagging requests that are near or beyond that target, updated in real time. This is a measurable improvement in monitoring accuracy and speed compared to the manual or purely workflow-based tracking described in the reviewed literature.

4. Local Optimization

Definition applied: Adapting a solution that works elsewhere to fit a specific local problem or constraint.

- A recurring limitation across all four studies is that the developed systems were built for the specific resources, procedures, and scale of one local government unit, which limits reuse elsewhere. BARANGAYAN adapts widely available, low-cost tools — Leaflet.js and OpenStreetMap for mapping, and Supabase's free service tier for the backend — into a system specifically designed to work within the limited budgets of Philippine barangays, which typically cannot afford metered mapping platforms or dedicated IT infrastructure. This reflects a deliberate adaptation of existing, general-purpose tools to the specific resource limitations identified as a common challenge among barangays.

5. Unique Dataset

Definition applied: Testing an established idea using new, previously unstudied data.

- This element applies only in a limited way to BARANGAYAN. The pilot deployment at Barangay Ampid I, San Mateo, Rizal, would generate new, barangay-specific data on document requests, incident reports, and processing times that has not been examined in the reviewed studies, since each of those studies collected data from a different local government unit (Calamba, Los Baños, Paracale, and Barangay Yuson). However, this is a secondary point and should not be presented as a main novelty claim, since the data itself is not the primary contribution of the project.

Summary Statement for the Proposal

Based on this comparison, BARANGAYAN's novelty should be presented in the following order of strength:

1. **Primary novelty:** Feature integration — combining document processing, incident mapping, emergency coordination, and payment into one platform, where the reviewed literature shows these as separate systems.
2. **Secondary novelty:** A configurable design that other barangays can adopt without redevelopment, addressing the single-barangay limitation found in REPORTIT, LBigayAksyon, and Webyu.
3. **Supporting novelty:** Automated, real-time compliance tracking, which improves on the manual or workflow-only monitoring described in the Paracale and Webyu studies.
4. **Supporting novelty:** Use of free, low-cost tools suited to the resource limitations common among barangays.

Novelty of the Project (Using Existing Algorithm)

The novelty of the project lies in its system design and feature integration, as discussed in the Novelty section of this proposal. This point should be stated directly in the paper, since claiming an original algorithm without one

presents a greater risk during the defense than acknowledging that the project applies existing, well-established algorithms to the barangay setting.

1. Automated Service Tracking Module

- Algorithm to cite: Finite State Machine (FSM), combined with threshold-based expiration logic.
- Each request moves through a fixed sequence of states — submitted, in progress, and resolved — which follows the structure of a Finite State Machine. The system flags a request once the elapsed time passes its assigned target, following the same logic used in cache expiration systems, where a stored record expires after a set time limit. This function runs on Supabase through a scheduled Postgres function, which checks the elapsed time of each request against its target.

2. Community Incident Reporting and Mapping Module (location verification)

- Algorithm to cite: Point-in-Polygon Algorithm (Ray Casting).
- Determining whether a resident's device location falls within a barangay's boundary is a computational geometry problem with an established solution. A line is drawn from the point in question, and the number of times this line crosses the boundary determines whether the point lies inside or outside the area. This method applies directly to the Leaflet.js and OpenStreetMap mapping stack used in the system, since boundary polygons are standard OpenStreetMap data.

3. Community Incident Reporting and Mapping Module (map display)

- Algorithm to cite: Grid-Based Point Clustering, as implemented in the Leaflet.markercluster plugin.
- Incidents are grouped on the map according to category and resolution status. This grouping is achieved through grid-based clustering, an established and published technique already built into the Leaflet ecosystem used by the system.

4. Digital Payment Module and QR Code Check-In (Emergency Module)

- Standard to cite: ISO/IEC 18004 (QR Code Standard), with Reed-Solomon Error Correction.
- Both the PayMongo QR Ph payment code and the evacuation check-in QR code rely on this same standard, which allows a QR code to remain scannable even if part of it is damaged or obstructed.

5. Automated Service Tracking Module and Administrative Analytics Dashboard (real-time updates)

- Pattern to cite: Publish-Subscribe (Pub/Sub) Messaging, using PostgreSQL's Write-Ahead Log (WAL).
- Supabase Realtime, identified in the proposal as the platform's real-time engine, functions by reading PostgreSQL's Write-Ahead Log to detect changes in database rows and pushing these changes to subscribed clients through WebSockets. This is a standard pattern in distributed systems rather than custom logic developed specifically for this project.

6. Account, Security, and Privacy Compliance

- Standard to cite: Role-Based Access Control (RBAC), as formally defined by NIST, and TOTP/HOTP (RFC 6238 / RFC 4226) for the one-time password feature.
- Supabase enforces role-based access control through Row-Level Security policies at the database level, and its one-time password feature is built on one of these two published standards.

7. Emergency and DRRM Information Module (evacuation center routing)

- Algorithm to cite: **Haversine Formula**, combined with **OSRM (Open Source Routing Machine) using Contraction Hierarchies**, a preprocessing-based extension of **Dijkstra's Algorithm**.
- This feature applies a two-step approach. First, the Haversine Formula calculates the great-circle distance between the resident's current device location and each evacuation center, identifying the nearest center or centers through a nearest-neighbor comparison. Second, the shortlisted centers are passed to the OSRM routing engine through the Leaflet Routing Machine plugin, which computes the actual road-following route using Contraction Hierarchies and displays the resulting route directly on the map, built on the Leaflet.js and OpenStreetMap stack already used elsewhere in the system.

8. Community Incident Reporting and Mapping Module (crowdsourced confirmation)

- Algorithm to cite: Crowdsourced Corroboration / Voting-Based Trust Scoring.
- When multiple residents confirm the same incident report, the system tallies these confirmations as a simple trust signal, following a pattern used in applications such as Waze for traffic and hazard reporting. This is an established crowdsourcing technique rather than one developed specifically for this project, and is implemented as a confirmation count linked to each incident report, limited to one confirmation per resident account.

9. Automated Service Tracking Module (trend-based flagging)

- Algorithm to cite: Rolling Average / Moving Average Trend Analysis.
- In addition to flagging a request once it individually exceeds its target, the system computes a rolling average of actual completion times per document type over a recent period, such as the last thirty days, and compares this average against the configured target. This is a standard time-series monitoring technique rather than a machine learning model, and should be described as rule-based trend detection rather than predictive artificial intelligence in the paper.

10. Trash and Illegal Dumping Priority Mapping Module

- Algorithm to cite: Time-Decay Weighted Frequency Scoring.
- Trash-related and illegal-dumping reports are grouped by area, and each area is given a score based on how many reports it has received and how recent those reports are, with older reports counting for less. This follows the same principle as time-decay ranking methods used in general computing (for example, ranking systems that lower an item's score as time passes so that older activity does not outweigh recent activity). This is implemented as a rule-based scoring method rather than a trained machine learning model, since a six-month, single-barangay pilot is unlikely to generate enough historical reports to train and validate a model such as Random Forest with confidence. This approach should be described plainly as rule-based priority scoring, not predictive artificial intelligence, in the paper.

11. Medical and Vaccination Drive Module

- Algorithm to cite: Priority Queue with Weighted Eligibility Scoring, combined with Atomic Transaction Concurrency Control for the stock decrement.
- Each registrant gets a computed score from defined weights (e.g., senior citizen +3, PWD +2, comorbidity +2, general public +0). Registrants are sorted into a priority queue (a standard data structure heap-based priority queue), and when stock is confirmed, the system allocates top-down until stock is exhausted, placing the remainder on a waitlist. To prevent two residents from being simultaneously assigned the

same last unit of stock, each allocation runs as an atomic Postgres transaction on Supabase (row-level locking / a Postgres function), a standard concurrency-control technique used in inventory and ticket-booking systems.

Summary table for direct use in your Technical Background section:

Module	Existing Algorithm/Standard
Automated Service Tracking Module	Finite State Machine + threshold-based expiration
Location verification (Incident Module)	Point-in-Polygon (Ray Casting)
Incident map display (Incident Module)	Grid-Based Point Clustering (Leaflet.markercluster)
Digital Payment Module / QR check-in	ISO/IEC 18004 with Reed-Solomon Error Correction
Real-time updates (Tracking Module, Dashboard)	Publish-Subscribe via PostgreSQL WAL
Account, Security, and Privacy Compliance	RBAC (NIST model); TOTP/HOTP for OTP
Evacuation center routing (Emergency and DRRM Information Module)	Haversine Formula + OSRM (Contraction Hierarchies / Dijkstra's Algorithm)
Incident confirmation (Incident Module)	Crowdsourced Corroboration / Voting-Based Trust Scoring
Trend-based flagging (Tracking Module)	Rolling Average / Moving Average Trend Analysis
Trash and Illegal Dumping Priority Mapping Module	Time-Decay Weighted Frequency Scoring

Online References Supporting the Novelty of BARANGAYAN (2022–2026)

1. Supporting the Gap: Existing Systems Are Single-Function or Single-Barangay

These sources confirm that current Philippine barangay digitization projects remain limited to one function and one locality, which is the exact gap your Feature Integration and Configurable Design claims respond to.

- Gabo, D. S., & Velasco, M. D. (2023). *Digitalization in Local Government Units in the Philippines: Challenges and Opportunities.* 2023 8th International Conference on Information Technology Research (ICITR).
<https://ieeexplore.ieee.org/document/10468134>
- SIEEE. (2023). *Digital Transformation in Local Governance: A Case Study of Barangay-Level Technology Adoption in the Philippines.* IEEE.
<https://ieeexplore.ieee.org/document/10912845>
- Ballaran, M. J. G., et al. (2023). *Development and Evaluation of Web-Based Barangay Profiling and Issuance System.* World Journal of Advanced Research and Reviews, 20(03), 060–072.
<https://wjarr.com/sites/default/files/WJARR-2023-2404.pdf>
- Docu-Go: The Development and Assessment of a Web-Based Barangay Document Requesting System. (2023). International Journal of Information Technology & Computer Engineering.
<https://journal.hmjournals.com/index.php/IJITC/article/view/2382>
- Roslan, M. A. A., & Haron, H. (2024). *Mobile Community App: Developing Barangay Management Systems.*
https://www.researchgate.net/publication/399120939_Mobile_Community_App_Developing_Barangay_Management_Systems

Each of these confirms your paper's claim: current systems handle only document processing or records management for a single barangay, with no combined incident-mapping, emergency, or payment functions.

2. Supporting the Digital Payment Module (PayMongo + QR Ph)

These are official regulatory sources confirming that QR Ph is the mandated, interoperable national standard your payment design correctly relies on.

- Bangko Sentral ng Pilipinas. (2023). *Memorandum No. M-2023-005: Adoption of the National QR Code Standard (QR Ph).*
- GMA News Online. (2023). *BSP mandates full adoption of Philippine QR Code standard by July 1.*

<https://www.bsp.gov.ph/Regulations/Issuances/2023/M-2023-005.pdf>

<https://www.gmanetwork.com/news/money/economy/863474/bsp-mandates-full-adoption-of-philippine-qr-code-standard-by-july-1/story/>

These support the claim in your Technology Stack section that QR Ph is a government-backed, interoperable standard, not a proprietary or optional feature.

3. Supporting the Location Verification Feature (Point-in-Polygon Geofencing)

These confirm the algorithm named in the earlier technical discussion is an established, published method, not an invention of this project.

- *Implementation of Geo-fencing to Monitor a Specific Target Using Point-in-Polygon Algorithm.* (2022). International Research Journal of Engineering and Technology (IRJET), 9(12).
<https://www.slideshare.net/irjetjournal/implementation-of-geofencing-to-monitor-a-specific-target-using-point-in-polygon-algorithm>

- *Dynamic Polygon Geofencing for Spatial Validation in Campus Attendance Monitoring.* (2025). Journal of Engineering & Technological Advances.
<https://jeta.segi.edu.my/index.php/segi/article/view/144>

These support the claim that your Location Verification feature uses a recognized method with known accuracy limitations, matching the "soft check" design already described in your proposal.

4. Supporting the Automated Service Tracking Module (Real-Time Updates)

This is the official technical documentation for Supabase Realtime, confirming the Publish-Subscribe/WAL-based mechanism your SLA tracking depends on.

- Supabase. *Realtime Architecture.* Supabase Docs.
<https://supabase.com/docs/guides/realtime/architecture>

This is the primary source to cite for the real-time synchronization claim in your Project Description and Technology Stack sections, since it is Supabase's own official documentation.

5. Supporting the Incident Map Display (Clustering)

- Leaflet. *Leaflet.markercluster: Marker Clustering Plugin for Leaflet.* Official repository.
<https://github.com/Leaflet/Leaflet.markercluster>

6. Supporting the Trash and Illegal Dumping Management Mapping Module

- This study addresses improper solid waste management as a major concern in the Philippines, noting that Metro Manila alone affects nearly twelve million people, and responds by promoting proper solid waste management at the barangay level through a mobile application framework. Fabito, B. S., et al. Perazuhan: A Mobile Application for Solid Waste Micro-Management Framework. IEEE Conference Publication.
<https://ieeexplore.ieee.org/document/8896301/>

7. Supporting the Emergency and DRRM Evacuation Routing Feature

These sources confirm that both algorithms applied in the evacuation routing feature are established, previously studied techniques rather than inventions of this study.

- Lagman, A. C., Angel, R. D., Ramos, R. F., Perez, M. R. L., & Maaliw, R. R. (2022). *I-Respond: Mobile Application for Emergency Response Using Dijkstra's Algorithm Shortest Path*. 2022 IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), 1–5. <https://doi.org/10.1109/HNICEM57413.2022.10109620>

This IEEE conference paper, developed in the Philippine setting, applies Dijkstra's Algorithm to compute shortest paths for emergency response and was rated 4.52 out of 5 by respondents using the ISO 9126 evaluation tool — directly supporting the use of Dijkstra-based shortest-path computation for Philippine disaster and emergency response applications.

- Lucero, H. R., Lewis, K. P., & Fernando Jr., P. G. (2025). *Healthcare Services Locator with Web Mapping and Chatbot Utilizing the Haversine Distance Metrics Algorithm*. World Education Connect Multidisciplinary E-Publication.

This Philippine study applies the Haversine formula to identify the nearest healthcare facility to a user's current GPS location on an Android-based application, tested within a barangay in Quezon City, and reports low relative error compared against alternative distance formulas — directly supporting the use of the Haversine formula for nearest-facility determination in a Philippine barangay context.

8. Supporting the Medical and Vaccination Drive Module

These sources confirm that digital vaccination/immunization management is an active, government-recognized priority in the Philippines, and that self-registration/scheduling platforms measurably improve vaccination uptake — supporting the rationale for a resident-facing registration and priority-queue feature rather than a purely walk-in model.

- Department of Health, UNICEF Philippines, & Government of Japan. (2023). *DigiVacc: Digital Immunization System for Children in the Philippines* (VaccTrace and VaccCheck applications). <https://www.unicef.org/philippines/press-releases/department-health-unicef-and-government-japan-launch-digital-immunization-system>
- Cabrera, L. et al. (2022). *COVID-19 Vaccination Drive in a Low-Volume Primary Care Clinic: Challenges & Lessons Learned in Using Homegrown Self-Scheduling Web-Based Mobile Platforms*. PMC. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9319099/>

These support the claim in your Problem Statement / rationale that (a) government agencies (DOH, UNICEF) are actively pushing digital immunization tracking down to the LGU level — DigiVacc was deployed across 13 LGUs specifically to connect caregivers and health workers and (b) self-registration and scheduling tools have a documented effect on uptake, since the primary-care clinic study reported 51% vaccination uptake using a self-scheduling web/SMS platform versus 36–39% uptake in the surrounding town, county, and state during the same period.

Online References Supporting the Problem Statement of BARANGAYAN (2022–2026)

- Philippine Information Agency. (2025). DILG pushes full digital shift in barangays through nationwide LGUSS-BIMS adoption. <https://pia.gov.ph/press-release/dilg-pushes-full-digital-shift-in-barangays-through-nationwide-lguss-bims-adoption/>
- Inquirer.net. (2025). DILG wants more barangays to use digital info management system. <https://newsinfo.inquirer.net/2079736/dilg-wants-more-barangays-to-use-digital-info-management-system>
- National Police Clearance. (2026). Barangay Clearance Online | Getting Barangay Clearance in the Philippines. <https://policeclearanceph.ph/barangay-clearance/>
- International Journal of Research and Innovation in Social Science. (2025). Disaster Risk Reduction and Management Programs (citing GSMA, 2022; PreventionWeb/UNDRR, 2023–2025). https://rsisinternational.org/journals/ijrsi/uploads/vol12-iss10-pg947-954-202511_pdf.pdf
- Padayon UP Public Service Office. (2025). 5 Apps, Tools Developed in UP Everyone Should Know About (and Use) this Rainy Season. <https://publicservice.up.edu.ph/5-apps-tools-developed-in-up-everyone-should-know-about-and-use-this-rainy-season/>
- Aliling, J. T., Santa Ana, B. B., & Soberano, K. T. (2025). Digitalizing Governance: A Transformation on the Processes of One Community in the Philippines. *International Journal of Multidisciplinary Research and Analysis*, 8(06). <https://www.ijmra.in/v8i6/Doc/68.pdf>
- Development of BRRMS: A Barangay Records Management System for Enhanced Services. (2024). <https://www.studocu.com/ph/document/aisat-college-dasmarinas/bs-computer-science/development-of-brrms-a-barangay-records-management-system-for-enhanced-services/135501111>

Technology Stack

- **React Native and Expo:** are used to build the resident-facing mobile application from a single codebase for both Android and iOS devices, without additional licensing costs.
- **Supabase:** provides the database, authentication, real-time data synchronization, server-side functions, and file storage used by both the resident application and the administrative dashboard. Its free service tier is sufficient for a project of this scale.
- **Leaflet.js and OpenStreetMap:** are used to display the incident and evacuation-center maps within the administrative dashboard. These tools are free to use and do not require an access key. Within the mobile application, the map is displayed using an embedded web view.
- **PayMongo and QR Ph:** are used for processing document and service payments. During development, the payment gateway operates in a test mode that does not incur any cost.